

$$1(a) \quad X_t = 1 + 2t + aX_{t-1} + Z_t$$

$$Y_t = \nabla^2 X_t$$

$$= \nabla(\nabla X_t)$$

$$= \nabla[(1 + 2t + aX_{t-1} + Z_t) - (1 + 2(t-1) + aX_{t-2} + Z_{t-1})]$$

$$= \nabla[2 + aX_{t-1} - aX_{t-2} + Z_t - Z_{t-1}]$$

$$= (2 + aX_{t-1} - aX_{t-2} + Z_t - Z_{t-1}) - (2 + aX_{t-2} - aX_{t-3} + Z_{t-1} - Z_{t-2})$$

$$= a(X_{t-1} - 2X_{t-2} + X_{t-3}) + Z_t - \cancel{Z_{t-1}}^2 + Z_{t-2}$$

$$= aY_{t-1} + Z_t - 2Z_{t-1} + Z_{t-2}$$

$$(b) \quad (1 - aB)Y_t = Z_t - 2Z_{t-1} + Z_{t-2} \\ = (1 - 2B + B^2)Z_t$$

The process is (weakly) stationary if roots of $1 - az = 0$ lie outside the unit circle

The root is $z = \frac{1}{a} \therefore$ (weakly) stationary if $|a| < 1$.

(c) Given

$$(1 - aB)Y_t = (1 - 2B + B^2)Z_t$$

$$\Rightarrow Y_t = \frac{(1 - 2B + B^2)Z_t}{1 - aB} \quad (*)$$

$$\text{Let } f(B) = (1 - aB)^{-1}$$

$$f'(B) = a(1 - aB)^{-2}$$

$$f''(B) = 2a^2(1 - aB)^{-3}$$

Generally, if $f^{(k)}(B)$ is the k^{th} derivative of $f(B)$ then

$$f^{(k)}(B) = k! a^k (1 - aB)^{-(k+1)}$$

The Taylor series expansion of $f(B)$ about 0 is

$$f(B) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0) B^k}{k!}$$

$$= \sum_{k=0}^{\infty} a^k B^k.$$

$\therefore$ in $(*)$

$$Y_t = (1 - 2B + B^2) \sum_{k=0}^{\infty} a^k B^k Z_t$$

$$= \sum_{k=0}^{\infty} a^k B^k Z_t - 2 \sum_{k=0}^{\infty} a^k B^{k+1} Z_t + \sum_{k=0}^{\infty} a^k B^{k+2} Z_t$$

$$= Z_t + (a-2) Z_{t-1} + \frac{(a-1)^2}{a^2} \sum_{k=2}^{\infty} a^k Z_{t-k}.$$

(d) $a = \frac{1}{2}$

Consider

$$\begin{aligned} \gamma(h) &= \text{Cov}(Y_t, Y_{t+h}) \\ &= E(Y_t Y_{t+h}) - E(Y_t) E(Y_{t+h}) \\ &= E(Y_t Y_{t+h}) \end{aligned}$$

because $E(Y_t) = 0$ for all t .

Then

$$\begin{aligned} \gamma(h) &= E(Y_t Y_{t+h}) \\ &= E \left[\left(\sum_{j=0}^{\infty} b_j Z_{t-j} \right) \left(\sum_{k=0}^{\infty} b_k Z_{t+h-k} \right) \right] \\ &= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} b_j b_k E(Z_{t-j} Z_{t+h-k}) \end{aligned}$$

Since $\{Z_t\}$ is a white noise process

$$E(Z_{t-j} Z_{t+h-k}) = 0 \text{ except where}$$

$$-j = h - k \Rightarrow k = j + h$$

Then

$$\begin{aligned}\gamma(h) &= \sum_{j=0}^{\infty} b_j b_{j+h} E(z_{t-j}^2) \\ &= \sigma_z^2 \sum_{j=0}^{\infty} b_j b_{j+h}\end{aligned}$$

Note $b_0 = 1$

$$b_1 = a - 2 = -\frac{3}{2}$$

$$b_j = \frac{(a-1)^2}{a^2} a^j = \left(\frac{1}{2}\right)^j \quad j \geq 2.$$

For $h=0$:

$$\gamma(0) = \sigma_z^2 \sum_{j=0}^{\infty} b_j^2$$

$$= \sigma_z^2 \left(1 + \left(-\frac{3}{2}\right)^2 + \sum_{j=2}^{\infty} \left(\frac{1}{2}\right)^{2j} \right)$$

$$= \sigma_z^2 \left(\frac{13}{4} + \frac{1}{1 - \frac{1}{4}} \cdot \frac{1}{16} \right)$$

$$= \sigma_z^2 \left(\frac{13}{4} + \frac{1}{12} \right)$$

$$= \sigma_z^2 \frac{10}{3}.$$

Geometric series.

$$\gamma(h) = \begin{cases} \frac{10}{3} \sigma_z^2 & h=0; \\ -\frac{11}{6} \sigma_z^2 & |h|=1; \\ \frac{\sigma_z^2}{3} \left(\frac{1}{2}\right)^{|h|} & |h| \geq 2. \end{cases}$$

2. (a)

Consider

$$\begin{aligned} E(X_t) &= 150 + 0.4 E(X_{t-1}) + E(Z_t) \\ &\quad - 0.5 E(Z_{t-1}) \end{aligned}$$

$$\Rightarrow E(X_t) = 150 + 0.4 E(X_{t-1})$$

$$\Rightarrow \mu = 150 + 0.4 \mu$$

$$\Rightarrow \mu = 250.$$

$$(b) \quad X_t = 150 + 0.4 X_{t-1} + Z_t - 0.5 Z_{t-1}$$

$$\text{Consider } Y_t = X_t - 250$$

$$\Rightarrow Y_t = 0.4 Y_{t-1} + Z_t - 0.5 Z_{t-1}$$

$$\Rightarrow (1 - 0.4B) Y_t = (1 - 0.5B) Z_t$$

Invertible if roots of

$1 - 0.5z = 0$ lie outside the unit circle

$z = 2 \therefore \{Y_t\}$ is invertible

$\Rightarrow \{X_t\}$ is invertible.

Note $Z_t = \frac{(1 - 0.4B) Y_t}{1 - 0.5B} \quad (*)$

Let $f(B) = 1 - 0.5B$

If $f^{(k)}(B)$ is the k^{th} derivative of $f(B)$

then

$$f^{(k)}(B) = k! 0.5^k (1 - 0.5B)^{-(k+1)}$$

$$\therefore f(B) = \sum_{k=0}^{\infty} 0.5^k B^k$$

Hence in $(*)$

$$Z_t = (1 - 0.4B) \sum_{k=0}^{\infty} 0.5^k B^k Y_t$$

$$= (1 - 0.4B) \sum_{k=0}^{\infty} 0.5^k Y_{t-k}$$

$$= Y_t + \sum_{k=1}^{\infty} (0.5 - 0.4) 0.5^{k-1} Y_{t-k}$$

$$= Y_t + 0.1 \sum_{k=1}^{\infty} 0.5^{k-1} Y_{t-k}$$

$$= (X_t - 250) + 0.1 \sum_{k=1}^{\infty} 0.5^{k-1} (X_{t-k} - 250)$$

$$(c) \quad Y_t - 0.4Y_{t-1} = Z_t - 0.5Z_{t-1}$$

Suppose the series is observed to time n .

We define

$$[Z_{n+k}] = \begin{cases} 0 & k > 0; \\ Z_{n+k} & k \leq 0. \end{cases}$$

Then

$$\hat{Y}_{n+k} - 0.4\hat{Y}_{n+k-1} = [Z_{n+k}] - 0.5[Z_{n+k-1}]$$

Observed Y values

t	2021	2022	2023	2024
Y_t	-10	18	4	22

$k = 1$ (2025)

$$\hat{Y}_{2025} - 0.4Y_{2024} = [Z_{2025}] - 0.5[Z_{2024}]$$

$$\hat{Y}_{2025} = 0.4Y_{2024} - 0.5Z_{2024}$$

$$Z_{2024} = Y_{2024} + 0.1 \sum_{k=1}^{\infty} 0.5^{k-1} Y_{2024-k}$$

$$= 23.05$$

$$\text{Finally } \hat{Y}_{2025} = (0.4 \times 22) - 0.5 \times 23.05 = -2.725$$

$$\begin{aligned}\hat{X}_{2025} &= \hat{Y}_{2025} + 250 \\ &= 247.725\end{aligned}$$

3. (a)

Time plot $\Rightarrow$ process is stationary with a constant mean, and constant variance.

ACF plot $\Rightarrow$ spike at lag 1 then cuts off to zero. (perhaps MA(1)).

PACF plot $\Rightarrow$ gradually declines to zero, suggesting a pure MA process.

$\therefore$ An appropriate model is

$$V_t = Z_t + \theta Z_{t-1}.$$

$$\sigma_z^2 : \cancel{\text{Var}(Z_t)} = (1 + \theta B) Z_t$$

For $\hat{\theta}$:

The autocovariance generating f'n is

$$\begin{aligned} \Gamma(z) &= \sigma_z^2 (1 + \theta z)(1 + \theta z^{-1}) \\ &= \sigma_z^2 (1 + \theta^2) + \sigma_z^2 \theta z + \sigma_z^2 \theta z^{-1} \end{aligned}$$

$\therefore$ ACF at lag 1 is $\frac{\theta}{1 + \theta^2}$

$$\therefore \text{Set } \frac{\theta}{1 + \theta^2} = 0.4 \quad (\text{ACF plot})$$

$$\theta = 0.5 \text{ or } 2. \quad \text{Choose } \hat{\theta} = 0.5 \quad (\text{invertibility})$$

$$\hat{\sigma}_z^2 :$$

$$\text{Var}(V_t) = \sigma_z^2 (1 + \theta^2)$$

$$\text{Set } 2 \times \sqrt{\text{Var}(V_t)} = 1$$

$$\Rightarrow 2 \sigma_z^2 \sqrt{1 + \hat{\theta}^2} = 1$$

$$\Rightarrow \hat{\sigma}_z^2 = \frac{1}{2\sqrt{1 + \hat{\theta}^2}}$$

$$\hat{\sigma}_z^2 = \frac{1}{2\sqrt{1 + \frac{1}{4}}}$$

$$= \frac{1}{2\sqrt{\frac{5}{4}}} = \frac{1}{\sqrt{5}}$$

(b) The incorrect model is

$$X_t = 0.4 X_{t-1} + Z'_t - 0.5 Z'_{t-1} + 0.06 Z'_{t-2}$$

$$\Rightarrow (1 - 0.4B) X_t = (1 - 0.5B + 0.06B^2) Z'_t$$

$$\Rightarrow X_t = [\phi_0(B)]^{-1} \theta_0(B) Z'_t \quad (*)$$

$$\text{where } \phi_0(B) = 1 - 0.4B$$

$$\theta_0(B) = 1 - 0.5B + 0.06B^2$$

$$\text{We have } Z'_t = 0.5 Z'_{t-1} + Z_t$$

$$\Rightarrow (1 - 0.5B) Z'_t = Z_t$$

$$Z'_t = (1 - 0.5B)^{-1} Z_t$$

$\therefore \ln (*)$

$$X_t = [\phi_o(B)]^{-1} \theta_o(B) (1 - 0.5B)^{-1} Z_t$$

$$\Rightarrow (1 - 0.5B) \phi_o(B) X_t = \theta_o(B) Z_t$$

an ARMA(2,2) model.

$$(c) \left(1 - \frac{B}{4} - \frac{B^2}{8}\right) Y_t = Z_t$$

$$\Rightarrow Y_t = \left(1 - \frac{B}{4} - \frac{B^2}{8}\right)^{-1} Z_t$$

Then

$$U_t = Y_t + \frac{2}{3} Y_{t-1} + \frac{1}{24} Y_{t-2}$$

$$= \left(1 - \frac{B}{4} - \frac{B^2}{8}\right)^{-1} Z_t$$

$$+ \frac{2}{3} \left(1 - \frac{B}{4} - \frac{B^2}{8}\right)^{-1} Z_{t-1}$$

$$+ \frac{1}{24} \left(1 - \frac{B}{4} - \frac{B^2}{8}\right)^{-1} Z_{t-2}$$

$$\Rightarrow \left(1 - \frac{B}{4} - \frac{B^2}{8}\right) U_t = Z_t + \frac{2}{3} Z_{t-1} + \frac{1}{24} Z_{t-2}.$$

ARMA(2,2) process.

4(a)

$$\left(1 + \frac{B}{2} - \frac{3}{16}B^2\right) X_t = \left(1 - \frac{B}{4}\right) Z_t$$

$$\left(1 - \frac{B}{4}\right) \left(1 + \frac{3}{4}B\right) X_t = \left(1 - \frac{B}{4}\right) Z_t$$

$$\Rightarrow \left(1 + \frac{3B}{4}\right) X_t = Z_t$$

$$X_t = \left(1 - \frac{3B}{4}\right)^{-1} Z_t$$

$$= \alpha(B) Z_t$$

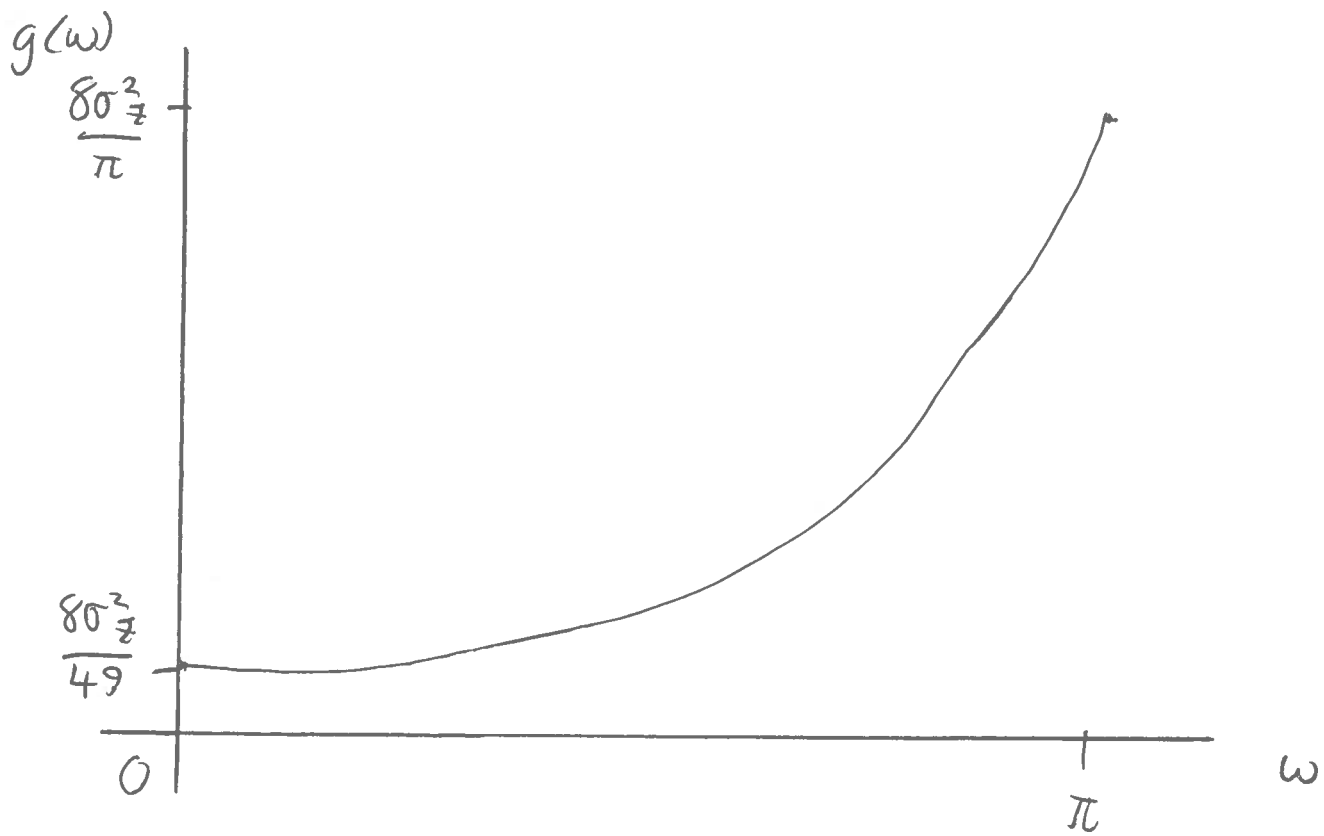
$$\text{where } \alpha(B) = \left(1 - \frac{3}{4}B\right)^{-1}$$

Spectral density f^n :

$$g(\omega) = \frac{\sigma_z^2}{2\pi} \alpha(e^{i\omega}) \alpha(e^{-i\omega})$$

$$= \frac{\sigma_z^2}{2\pi} \left[\frac{1}{\left(1 - \frac{3}{4}e^{i\omega}\right)\left(1 - \frac{3}{4}e^{-i\omega}\right)} \right]$$

$$= \frac{2\sigma_z^2}{\pi} \left[\frac{1}{\left(\frac{25}{4} + 6\cos\omega\right)} \right]$$



- High frequency behaviour
- Not a smooth process
- Rapid fluctuations about the mean

(c) Transfer function

$$h(\omega) = \sum_{j=-\infty}^{\infty} b_j e^{-i\omega j}$$

$$b_j = \begin{cases} 1 & j=0 \\ -\frac{9}{16} & j=2 \\ 0 & \text{o/w} \end{cases}$$

$$\therefore h(\omega) = \left(1 - \frac{9}{16} e^{-i2\omega} \right)$$

$$|h(\omega)|^2 = \left| 1 - \frac{9}{16} e^{-i2\omega} \right|^2$$

$$\begin{aligned}
 &= \left(1 - \frac{9}{16} e^{-i2\omega}\right) \left(1 - \frac{9}{16} e^{i2\omega}\right) \\
 &= 1 + \frac{81}{256} - \frac{9}{16} \underbrace{(e^{i2\omega} + e^{-i2\omega})}_{2\cos 2\omega} \\
 &= \left(\frac{25}{16} + \frac{3}{2} \cos \omega\right) \left(\frac{25}{16} - \frac{3}{2} \cos \omega\right)
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad g_y(\omega) &= |h(\omega)|^2 g_x(\omega) \\
 &= \left(\frac{25}{16} + \frac{3}{2} \cos \omega\right) \left(\frac{25}{16} - \frac{3}{2} \cos \omega\right) \\
 &\quad \times \frac{2\sigma_z^2}{\pi} \left[\frac{1}{\left(\frac{25}{4} + 6\cos \omega\right)} \right] \\
 &= \frac{\sigma_z^2}{2\pi} \left(\frac{25}{16} - \frac{3}{2} \cos \omega\right)
 \end{aligned}$$

